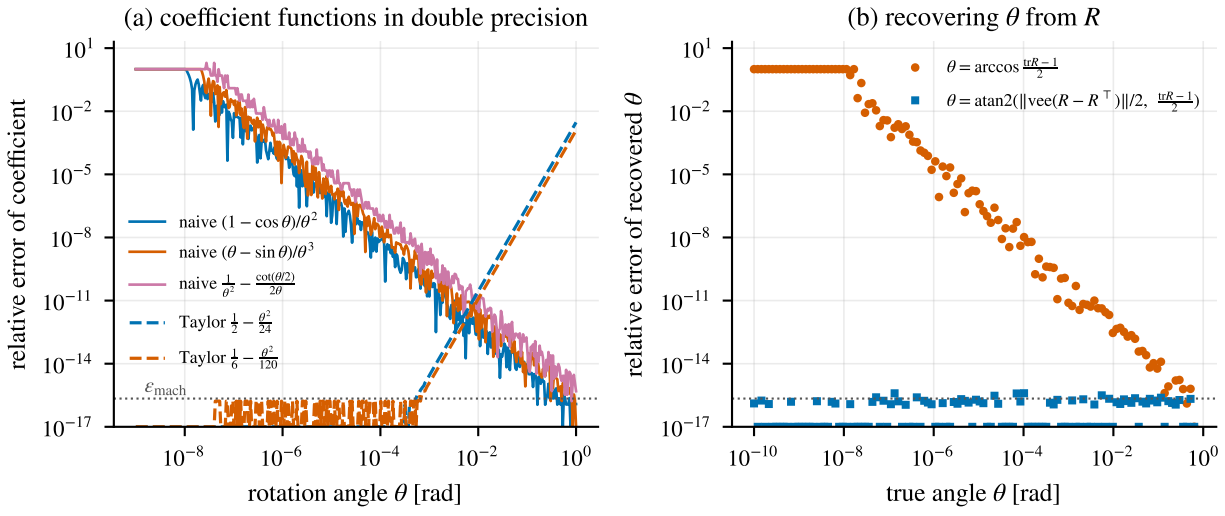
(a) coefficient functions in double precision
relative error of coefficient
rotation angle θ [rad]
naive $(1-\cos\theta)/\theta^2$
naive $(\theta-\sin\theta)/\theta^3$
naive $\frac{1}{\theta^2}-\frac{\cot(\theta/2)}{2\theta}$
Taylor $\frac{1}{2}-\frac{\theta^2}{24}$
Taylor $\frac{1}{6}-\frac{\theta^2}{120}$
$\varepsilon_{\text{mach}}$
(b) recovering θ from R
relative error of recovered θ
true angle θ [rad]
$\theta=\arccos\frac{\text{tr}R-1}{2}$
$\theta=\text{atan2}(\|\text{vee}(R-R^\top)\|/2,\ \frac{\text{tr}R-1}{2})$